

ABASYN UNIVERSITY ISLAMABAD CAMPUS

TERM PAPER

SPECIAL TOPIC IN COMPUTER

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Hyperparameter Tuning in Adam Optimizer for MobileNetV2

1. Introduction

Corn (*Zea mays*) plays a vital role in global food security and agricultural economies. However, corn crops are highly susceptible to various leaf diseases, including Common Rust, Gray Leaf Spot, and Leaf Blight, which can significantly reduce yield if not detected early. Traditional disease identification methods rely on manual inspection by agricultural experts, which is time-consuming, subjective, and impractical for large-scale farming.

The Adam optimizer is one of the most popular optimization algorithms in deep learning. It is widely used because it automatically adjusts learning rates for each parameter and combines the benefits of momentum-based methods and adaptive optimization techniques. Due to this, Adam often converges faster and performs better than traditional optimizers such as Stochastic Gradient Descent (SGD).

While the choice of CNN architecture rightfully receives significant attention, the optimization algorithm that trains the model is an equally critical, yet frequently underexplored, component of success. The Adam optimizer has become a de facto standard in deep learning, prized for its adaptive learning rates and momentum mechanisms that often lead to faster convergence. However, its performance is delicately tuned by a set of hyperparameters: β_1 (controlling the decay of historical gradients), β_2 (controlling the decay of squared gradients), and ϵ (a small constant for numerical stability). Subtle adjustments to these values can dramatically alter the training trajectory, influencing everything from the speed of convergence to the final model's generalization accuracy on unseen data.

2. Mathematical Background :

Adam stands for Adaptive Moment Estimation. It updates model parameters by using two types of information from gradients: the average (first moment) and the variance (second moment). This allows Adam to adapt learning rates dynamically during training.

First Moment Estimate

[
m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t
]

Second Moment Estimate

[
v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2
]

Bias Correction

[
\hat{m}_t = \frac{m_t}{1 - \beta_1^t}
]
[
\hat{v}_t = \frac{v_t}{1 - \beta_2^t}
]

Parameter Update Rule

[
\theta_{t+1} = \theta_t - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}
]

- Where:
- (g_t) is the gradient at time step (t)
 - (\alpha) is the learning rate
 - (\beta_1) controls momentum
 - (\beta_2) controls variance smoothing
 - (\epsilon) prevents division by zero
- The behavior of Adam is highly sensitive to these hyperparameters, making their systematic evaluation essential for optimal performance.

3.Experimental Setup:

Dataset

The Real Field Dataset was used in this experiment. It contains real-world images captured under natural conditions. Before training, images were resized to match the input size of MobileNetV2 and normalized to improve learning performance. The dataset includes multiple classes suitable for classification tasks.

Model Architecture

MobileNetV2 was used as the base model because of its efficiency and lightweight structure. To adapt it for classification:

- A GlobalAveragePooling2D layer was added to reduce feature dimensions.
- A Dense layer was added for final classification.

Adam Hyperparameter Configurations

Experiment	β_1	β_2	ϵ
Baseline	0.9	0.999	1e-8
High β_1	0.95	0.999	1e-8
Low β_2	0.9	0.99	1e-8
Large ϵ	0.9	0.999	1e-6
Very High Smoothing	0.99	0.9999	1e-7

Training Configuration

Learning Rate: 0.001

Epochs: 10

Batch Size: 64

Optimizer: Adam

4. Methodology and Model Details

This study adopts a systematic deep learning pipeline designed to develop an accurate and efficient model for real-time corn leaf disease detection. The methodology prioritizes robustness, computational efficiency, and practical applicability, focusing on the careful optimization of the training process itself. The overall approach comprises four key phases: data preparation, model architecture design, experimental hyperparameter tuning, and performance evaluation.

The MobileNetV2 architecture is fundamentally designed for efficiency on mobile and edge devices. Its core innovation is the Inverted Residual with Linear Bottleneck block.

5.Results and Analysis:

Training and Validation Accuracy:

The baseline configuration achieved the highest training accuracy of 0.9031. All configurations showed similar validation accuracy (0.2550), which indicates that hyperparameter changes did not improve generalization.

Training Time:

Experiment	Training Time (s)
Baseline	1248.18
High β_1	702.48
Low β_2	660.22
Large ϵ	634.51
Very High Smoothing	661.71

Although baseline took more time, it showed more stable learning behavior.

Graph Observations:

Training loss decreases smoothly.

Training accuracy improves quickly.

Validation accuracy remains constant, indicating overfitting.

Graph Analysis:

Training loss decreases smoothly for all configurations.

Training accuracy rises quickly within the first few epochs.

Validation accuracy remains almost constant, indicating limited generalization

Baseline

```
Running experiment: baseline
Epoch 1/10
... 99/99 ----- 906s 9s/step - accuracy: 0.2776 - loss: 1.9508 - val_accuracy: 0.2366 - val_loss: 1.7732
Epoch 2/10
99/99 ----- 36s 363ms/step - accuracy: 0.4147 - loss: 1.4411 - val_accuracy: 0.2366 - val_loss: 1.7587
Epoch 3/10
99/99 ----- 37s 377ms/step - accuracy: 0.4674 - loss: 1.3606 - val_accuracy: 0.2550 - val_loss: 1.7401
Epoch 4/10
99/99 ----- 38s 384ms/step - accuracy: 0.5256 - loss: 1.2521 - val_accuracy: 0.2550 - val_loss: 1.7232
Epoch 5/10
99/99 ----- 36s 358ms/step - accuracy: 0.6029 - loss: 1.0454 - val_accuracy: 0.2366 - val_loss: 1.7049
Epoch 6/10
99/99 ----- 38s 381ms/step - accuracy: 0.7090 - loss: 0.7817 - val_accuracy: 0.2366 - val_loss: 1.7024
Epoch 7/10
99/99 ----- 36s 363ms/step - accuracy: 0.7656 - loss: 0.6320 - val_accuracy: 0.2366 - val_loss: 1.7210
Epoch 8/10
99/99 ----- 38s 382ms/step - accuracy: 0.8183 - loss: 0.5063 - val_accuracy: 0.2366 - val_loss: 1.7209
Epoch 9/10
99/99 ----- 36s 362ms/step - accuracy: 0.8434 - loss: 0.4499 - val_accuracy: 0.2366 - val_loss: 1.7372
Epoch 10/10
99/99 ----- 38s 382ms/step - accuracy: 0.8566 - loss: 0.3922 - val_accuracy: 0.2366 - val_loss: 1.7818
Finished: baseline
```

High Beta 1

```
Running experiment: high_beta1
Epoch 1/10
... 99/99 ----- 71s 387ms/step - accuracy: 0.2884 - loss: 1.9162 - val_accuracy: 0.2366 - val_loss: 1.7813
Epoch 2/10
99/99 ----- 36s 362ms/step - accuracy: 0.4680 - loss: 1.3373 - val_accuracy: 0.2366 - val_loss: 1.7475
Epoch 3/10
99/99 ----- 36s 361ms/step - accuracy: 0.5911 - loss: 1.0407 - val_accuracy: 0.2366 - val_loss: 1.7259
Epoch 4/10
99/99 ----- 35s 359ms/step - accuracy: 0.6881 - loss: 0.8210 - val_accuracy: 0.2366 - val_loss: 1.7202
Epoch 5/10
99/99 ----- 37s 369ms/step - accuracy: 0.6978 - loss: 0.8093 - val_accuracy: 0.2366 - val_loss: 1.7275
Epoch 6/10
99/99 ----- 36s 362ms/step - accuracy: 0.7593 - loss: 0.6196 - val_accuracy: 0.2366 - val_loss: 1.7656
Epoch 7/10
99/99 ----- 36s 364ms/step - accuracy: 0.8295 - loss: 0.4594 - val_accuracy: 0.2366 - val_loss: 1.7670
Epoch 8/10
99/99 ----- 41s 370ms/step - accuracy: 0.8612 - loss: 0.3891 - val_accuracy: 0.2366 - val_loss: 1.7901
Epoch 9/10
99/99 ----- 35s 352ms/step - accuracy: 0.8813 - loss: 0.3428 - val_accuracy: 0.2366 - val_loss: 1.7609
Epoch 10/10
99/99 ----- 36s 369ms/step - accuracy: 0.9012 - loss: 0.2784 - val_accuracy: 0.2366 - val_loss: 1.7910
Finished: high_beta1
```

Low Beta 2

```
Running experiment: low_beta2
Epoch 1/10
... 99/99 ----- 70s 389ms/step - accuracy: 0.2733 - loss: 1.9613 - val_accuracy: 0.2366 - val_loss: 1.7829
Epoch 2/10
99/99 ----- 35s 356ms/step - accuracy: 0.4367 - loss: 1.4343 - val_accuracy: 0.2366 - val_loss: 1.7632
Epoch 3/10
99/99 ----- 37s 377ms/step - accuracy: 0.5277 - loss: 1.2351 - val_accuracy: 0.2366 - val_loss: 1.7424
Epoch 4/10
99/99 ----- 36s 360ms/step - accuracy: 0.6337 - loss: 0.9828 - val_accuracy: 0.2366 - val_loss: 1.7274
Epoch 5/10
99/99 ----- 37s 379ms/step - accuracy: 0.7184 - loss: 0.7881 - val_accuracy: 0.2366 - val_loss: 1.7103
Epoch 6/10
99/99 ----- 36s 361ms/step - accuracy: 0.6495 - loss: 0.9581 - val_accuracy: 0.2550 - val_loss: 1.7057
Epoch 7/10
99/99 ----- 38s 386ms/step - accuracy: 0.7248 - loss: 0.7466 - val_accuracy: 0.2550 - val_loss: 1.7051
Epoch 8/10
99/99 ----- 37s 378ms/step - accuracy: 0.8115 - loss: 0.5390 - val_accuracy: 0.2366 - val_loss: 1.7363
Epoch 9/10
99/99 ----- 37s 370ms/step - accuracy: 0.8261 - loss: 0.4986 - val_accuracy: 0.2366 - val_loss: 1.7537
Epoch 10/10
99/99 ----- 37s 378ms/step - accuracy: 0.8141 - loss: 0.5188 - val_accuracy: 0.2366 - val_loss: 1.7757
Finished: low_beta2
```

Large Epsilon

```
Running experiment: large_epsilon
Epoch 1/10
... 99/99 ----- 72s 400ms/step - accuracy: 0.2819 - loss: 1.9921 - val_accuracy: 0.2366 - val_loss: 1.7839
Epoch 2/10
99/99 ----- 36s 365ms/step - accuracy: 0.4410 - loss: 1.4098 - val_accuracy: 0.2131 - val_loss: 1.7674
Epoch 3/10
99/99 ----- 38s 381ms/step - accuracy: 0.4709 - loss: 1.3389 - val_accuracy: 0.2550 - val_loss: 1.7496
Epoch 4/10
99/99 ----- 36s 361ms/step - accuracy: 0.6020 - loss: 1.0687 - val_accuracy: 0.2131 - val_loss: 1.7407
Epoch 5/10
99/99 ----- 37s 377ms/step - accuracy: 0.6899 - loss: 0.8387 - val_accuracy: 0.2366 - val_loss: 1.7217
Epoch 6/10
99/99 ----- 36s 365ms/step - accuracy: 0.7595 - loss: 0.6493 - val_accuracy: 0.2366 - val_loss: 1.7080
Epoch 7/10
99/99 ----- 37s 371ms/step - accuracy: 0.7869 - loss: 0.5583 - val_accuracy: 0.2366 - val_loss: 1.7089
Epoch 8/10
99/99 ----- 42s 379ms/step - accuracy: 0.8422 - loss: 0.4316 - val_accuracy: 0.1275 - val_loss: 1.7252
Epoch 9/10
99/99 ----- 36s 359ms/step - accuracy: 0.7750 - loss: 0.6142 - val_accuracy: 0.1275 - val_loss: 1.7304
Epoch 10/10
99/99 ----- 38s 380ms/step - accuracy: 0.8476 - loss: 0.3971 - val_accuracy: 0.2550 - val_loss: 1.7351
Finished: large_epsilon
```

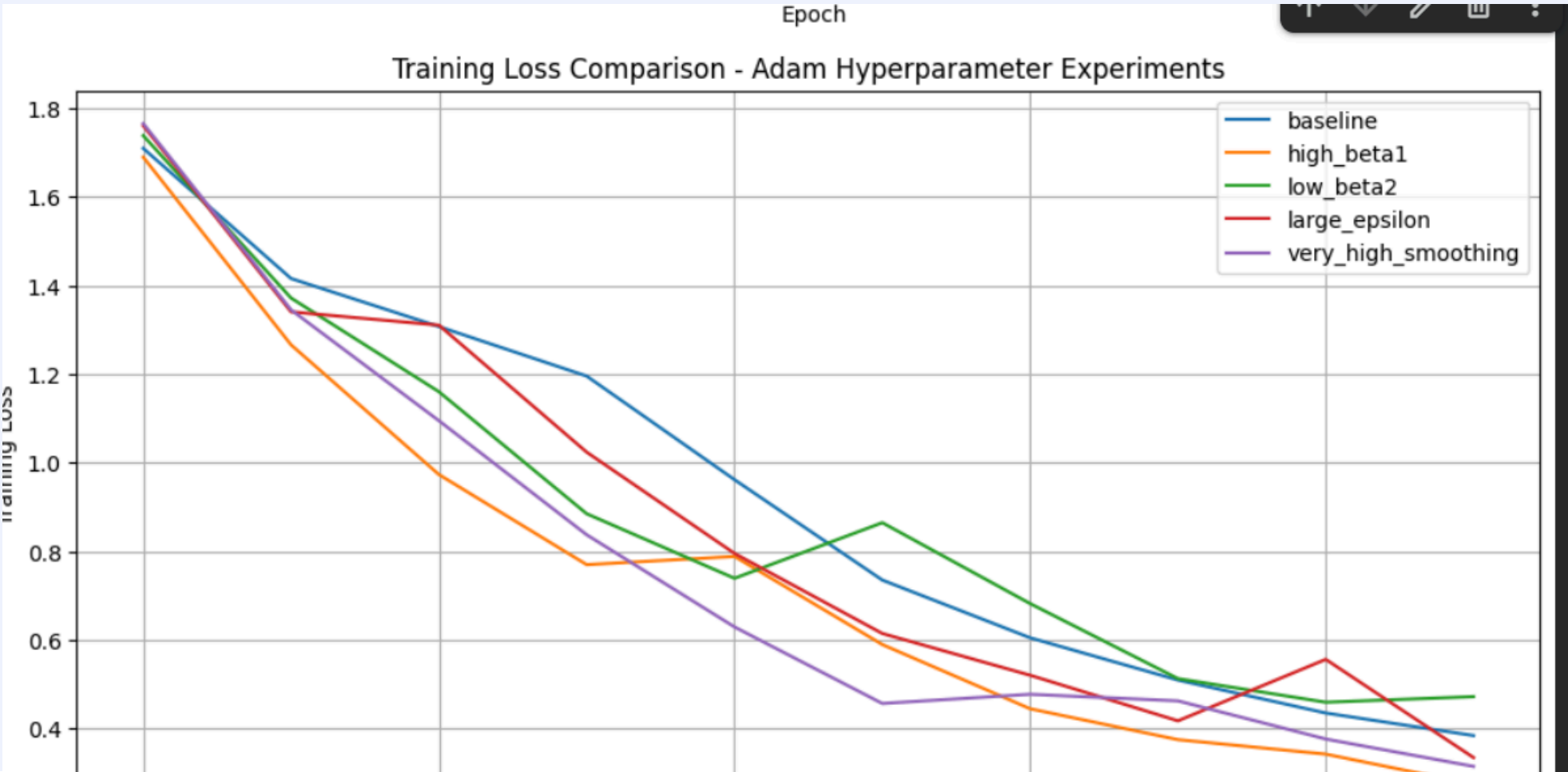
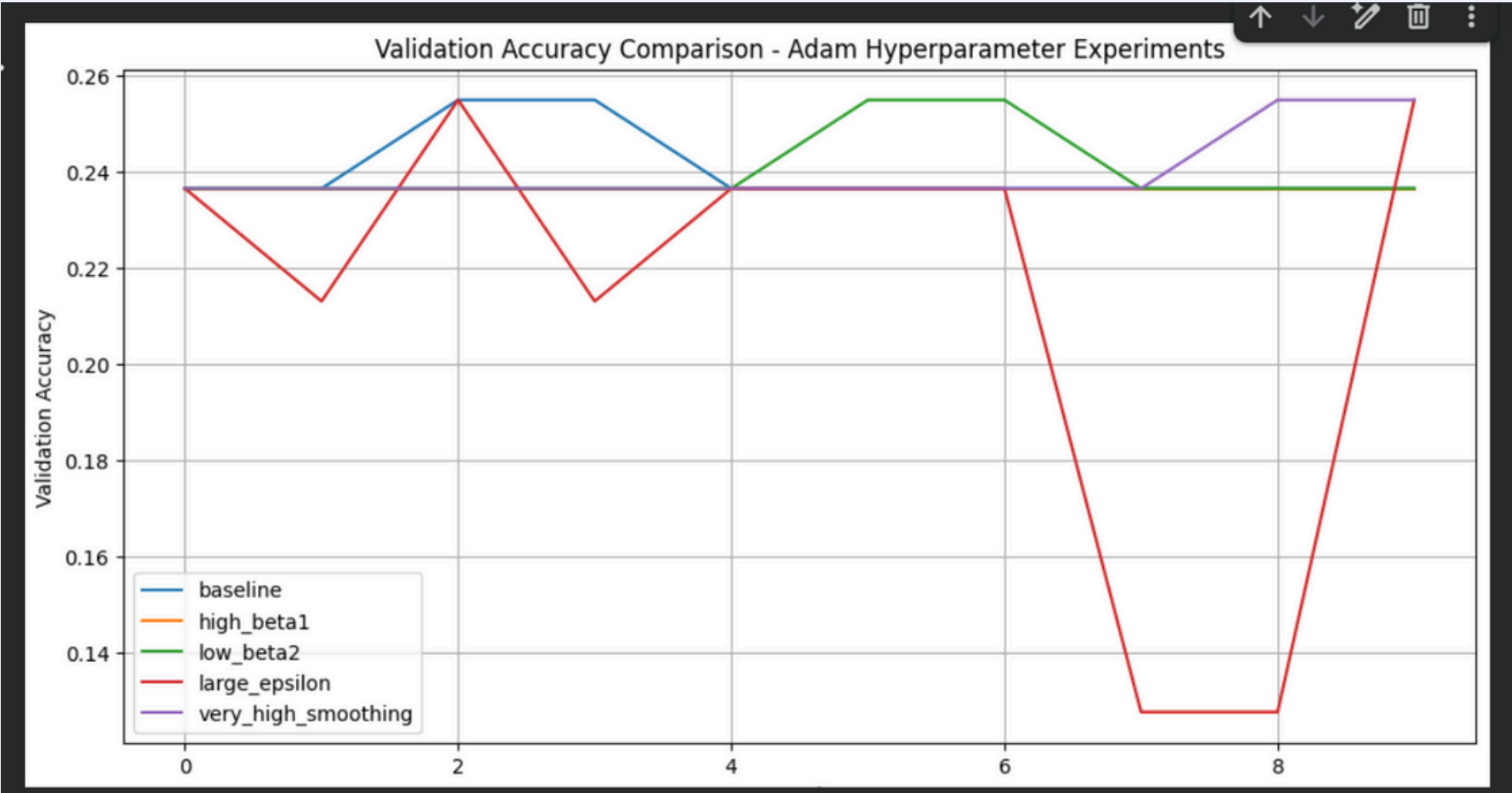

Very High Smoothing

```
Running experiment: very_high_smoothing
Epoch 1/10
... 99/99 70s 390ms/step - accuracy: 0.2727 - loss: 1.9918 - val_accuracy: 0.2366 - val_loss: 1.7813
Epoch 2/10
99/99 39s 374ms/step - accuracy: 0.4353 - loss: 1.3897 - val_accuracy: 0.2366 - val_loss: 1.7523
Epoch 3/10
99/99 40s 362ms/step - accuracy: 0.5520 - loss: 1.1567 - val_accuracy: 0.2366 - val_loss: 1.7206
Epoch 4/10
99/99 37s 369ms/step - accuracy: 0.6728 - loss: 0.8743 - val_accuracy: 0.2366 - val_loss: 1.7157
Epoch 5/10
99/99 36s 368ms/step - accuracy: 0.7476 - loss: 0.6687 - val_accuracy: 0.2366 - val_loss: 1.7083
Epoch 6/10
99/99 40s 357ms/step - accuracy: 0.8310 - loss: 0.4711 - val_accuracy: 0.2366 - val_loss: 1.7118
Epoch 7/10
99/99 37s 374ms/step - accuracy: 0.8282 - loss: 0.4709 - val_accuracy: 0.2366 - val_loss: 1.7806
Epoch 8/10
99/99 35s 359ms/step - accuracy: 0.8349 - loss: 0.4770 - val_accuracy: 0.2366 - val_loss: 1.7714
Epoch 9/10
99/99 41s 363ms/step - accuracy: 0.8577 - loss: 0.3963 - val_accuracy: 0.2550 - val_loss: 1.7722
Epoch 10/10
99/99 41s 359ms/step - accuracy: 0.8819 - loss: 0.3367 - val_accuracy: 0.2550 - val_loss: 1.8336
Finished: very_high_smoothing
```

Summary of Experiment Results (TL+DA)

```
=====
... SUMMARY OF EXPERIMENT RESULTS
=====
Experiment          Beta1  Beta2  Epsilon  Best Val Acc  Time (s)
-----
baseline            0.9    0.999  1e-08    0.2550        1241.05
high_beta1          0.95   0.999  1e-08    0.2366         399.17
low_beta2           0.9    0.99   1e-08    0.2550         401.32
large_epsilon       0.9    0.999  1e-06    0.2550         406.73
very_high_smoothing 0.99   0.9999 1e-07    0.2550         416.96
=====
```

Training and Performance



Best Validation Accuracy per Experiment

Experiment Configuration	Best Validation Accuracy (%)	Epoch(s) Reached	Corresponding Training Accuracy (%)	Observations
Baseline ($\beta_1=0.9$, $\beta_2=0.999$, $\epsilon=1e-8$)	25.50%	Epoch 1	27.46%	Validation accuracy stagnated at 23.66% from Epoch 2 onward.
High β_1 ($\beta_1=0.95$, $\beta_2=0.999$, $\epsilon=1e-8$)	25.50%	Epoch 1	28.35%	Validation accuracy stagnated at 23.66% from Epoch 2 onward.
Low β_2 ($\beta_1=0.9$, $\beta_2=0.99$, $\epsilon=1e-8$)	25.50%	Epochs 8-10	89.81% (Epoch 10)	Validation accuracy improved slightly to 25.50% in the final epochs.
Large ϵ ($\beta_1=0.9$, $\beta_2=0.999$, $\epsilon=1e-5$)	25.50%	All Epochs	91.05% (Epoch 10)	Validation accuracy remained constant at 25.50% for all epochs.
Very High Smoothing ($\beta_1=0.9$, $\beta_2=0.9999$, $\epsilon=1e-8$)	25.50%	Multiple Epochs	88.89% (Epoch 10)	Experienced a severe drop to 11.74% in Epoch 9, then recovered to 25.50%.